

# Clinical Effectiveness

## METIMMOX-1 Economic Evaluation

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# 1 Overview

This report presents clinical effectiveness results from the METIMMOX-1 trial, evaluating biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer.

The analysis uses **Firth-corrected Cox proportional hazards regression** with profile likelihood confidence intervals to assess treatment effect heterogeneity across two pre-treatment biomarkers identified by the causal DAG analysis:

- **CRP** (C-reactive protein): Low CRP (<5 mg/L) at baseline — a pre-treatment prognostic and potentially predictive marker
- **TMB/BRAF**: High tumor mutational burden ( 9 mut/MB) or BRAF mutation — a pre-treatment genomic marker

**Model specification is motivated by the causal DAG** (see `dag.qmd` and `dag_associations.qmd`).

The DAG encodes treatment (T) as randomized with no parents, so no confounding adjustment is needed for causal identification; Age and Sex are included as precision covariates. TLR (Tumor Lesion Reduction) is encoded as a post-randomization intermediate variable ( $T \rightarrow \text{TLR} \rightarrow \text{PFS}$ ): it is not a pre-treatment baseline characteristic and conditioning on it as a covariate would block part of the treatment effect pathway. TLR is therefore excluded from the primary survival models and examined separately as a descriptive endpoint.

The **unified DAG-informed model** is:

$$\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP} \times \text{Rx} + \text{TMB/BRAF} \times \text{Rx}$$

Ridge regression (L2-penalized Cox) is applied as a sensitivity analysis to assess the stability of the interaction estimates under shrinkage. TLR is reported in a separate exploratory section.

## 2 Methodological Notes

### 2.1 Software Packages

This analysis uses the following R packages:

- `coxphf` version 1.13.4: Firth-corrected Cox regression with profile likelihood CIs
- `survival` version 3.7.0: Cox regression and Schoenfeld residual diagnostics
- `glmnet` version 4.1.8: Ridge regression sensitivity analysis
- `vcd` version 1.4.13: Cramér’s V for biomarker correlation assessment

### 2.2 Firth-Corrected Cox Regression

We use Firth’s penalized maximum likelihood method for Cox regression, which addresses:

1. **Monotone likelihood**: When a covariate perfectly separates events from non-events
2. **Small sample bias**: Reduces bias in coefficient estimates with sparse data
3. **Unstable confidence intervals**: Profile likelihood CIs are more reliable than Wald-based CIs in small samples

Statistical inference is based on **penalized likelihood ratio tests (PLRT)**, which are more appropriate than Wald tests for testing treatment-biomarker interactions in this setting.

## 2.3 Ridge Regression Sensitivity Analysis

We use Ridge regression as a sensitivity analysis to assess the stability of our interaction estimates. Ridge regression (also called L2-penalized regression or Tikhonov regularization) belongs to the family of regularized or penalized regression methods. In survival analysis, this is implemented as a penalized Cox model.

### How Ridge regression works:

- Ridge adds a penalty term that discourages large coefficients. The bigger a coefficient tries to get, the more the model resists, pulling estimates toward zero (HR toward 1.0).
- Unlike LASSO ( $\alpha = 1$ ), which sets some coefficients exactly to zero and performs variable selection, Ridge ( $\alpha = 0$ ) shrinks all coefficients but retains all predictors.
- The degree of shrinkage is controlled by the tuning parameter  $\lambda$ , which is selected via **cross-validation (CV)**: the data are repeatedly split into training and validation sets, and the  $\lambda$  that produces the best predictive performance is chosen.

### Why not LASSO or Elastic Net?

We determined that LASSO and elastic net are **not appropriate** for this analysis:

1. **Correlated biomarkers:** LASSO arbitrarily selects one of correlated predictors while setting others to zero, which does not reflect the clinical question of comparing ALL biomarker strategies.
2. **Hierarchy principle violation:** LASSO may select interaction terms (e.g., `crp:Rx`) while setting the main effect (`crp`) to zero, producing models that are difficult to interpret clinically.
3. **Clinical objective:** Our goal is to estimate treatment effect heterogeneity across ALL biomarkers simultaneously, not to select a “winning” biomarker.

Elastic net ( $\alpha = 0.1$ – $0.3$ ) represents a compromise but does not fully address these concerns.

### Interpreting Ridge results:

Ridge regression is not replacing the Firth estimates—it provides a reality check. When Ridge substantially shrinks an interaction coefficient toward zero while Firth estimates a large effect, this indicates the effect is too unstable to trust for practical use. The CV-selected  $\lambda$  reflects how much regularization is needed for reasonable out-of-sample prediction given the sample size and collinearity structure.

## 3 Data Preparation

### 3.1 Sample Characteristics

### 3.2 Biomarker Correlations

**Note:** Cramér’s V interpretation:  $<0.1$  = Negligible,  $0.1$ – $0.3$  = Weak,  $>0.3$  = Moderate/Strong. The observed correlations are weak to negligible. While modest, any correlation among candidate biomarkers supports our decision not to use LASSO, which would arbitrarily select among correlated predictors.

Table 1: Sample Characteristics by Biomarker Status

| Characteristic                | Overall     | CRP         |             | TLR         |             | TMB/BRAF    |            |
|-------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|
|                               |             | Positive    | Negative    | Positive    | Negative    | Positive    | Negative   |
| Patients, n                   | 65          | 22          | 43          | 41          | 24          | 29          | 36         |
| <b>Demographics</b>           |             |             |             |             |             |             |            |
| Age, mean (SD)                | 64.0 (10.0) | 64.3 (10.3) | 63.8 (10.0) | 65.0 (9.9)  | 62.2 (10.2) | 64.6 (10.4) | 63.5 (9.8) |
| Male                          | 30 (46.2%)  | 12 (54.5%)  | 18 (41.9%)  | 19 (46.3%)  | 11 (45.8%)  | 14 (48.3%)  | 16 (44.4%) |
| Female                        | 35 (53.8%)  | 10 (45.5%)  | 25 (58.1%)  | 22 (53.7%)  | 13 (54.2%)  | 15 (51.7%)  | 20 (55.6%) |
| <b>Treatment arm</b>          |             |             |             |             |             |             |            |
| Control (FLOX alone)          | 29 (44.6%)  | 5 (22.7%)   | 24 (55.8%)  | 22 (53.7%)  | 7 (29.2%)   | 13 (44.8%)  | 16 (44.4%) |
| Experimental (FLOX/nivolumab) | 36 (55.4%)  | 17 (77.3%)  | 19 (44.2%)  | 19 (46.3%)  | 17 (70.8%)  | 16 (55.2%)  | 20 (55.6%) |
| <b>Outcome events</b>         |             |             |             |             |             |             |            |
| Deaths (OS events)            | 56 (86.2%)  | 16 (72.7%)  | 40 (93.0%)  | 32 (78.0%)  | 24 (100.0%) | 22 (75.9%)  | 34 (94.4%) |
| Progressions (PFS events)     | 48 (73.8%)  | 14 (63.6%)  | 34 (79.1%)  | 27 (65.9%)  | 21 (87.5%)  | 19 (65.5%)  | 29 (80.6%) |
| <b>Biomarker prevalence</b>   |             |             |             |             |             |             |            |
| CRP-positive (low CRP)        | 22 (33.8%)  | 22 (100.0%) | 0 (0.0%)    | 18 (43.9%)  | 4 (16.7%)   | 11 (37.9%)  | 11 (30.6%) |
| TLR-positive (early response) | 41 (63.1%)  | 18 (81.8%)  | 23 (53.5%)  | 41 (100.0%) | 0 (0.0%)    | 18 (62.1%)  | 23 (63.9%) |
| TMB/BRAF-positive             | 29 (44.6%)  | 11 (50.0%)  | 18 (41.9%)  | 18 (43.9%)  | 11 (45.8%)  | 29 (100.0%) | 0 (0.0%)   |

*Note:* Percentages within each column use the column subgroup as denominator. CRP/TLR/TMB-BRAF rows in the corresponding subgroup column equal 100% by definition.

Table 2: Biomarker Correlations (Cramér's V)

| Biomarker Pair  | Cramér's V | Interpretation |
|-----------------|------------|----------------|
| CRP vs TLR      | 0.278      | Weak           |
| CRP vs TMB/BRAF | 0.077      | Negligible     |
| TLR vs TMB/BRAF | 0.019      | Negligible     |

## 4 Proportional Hazards Assumption

### 4.1 Schoenfeld Residual Tests

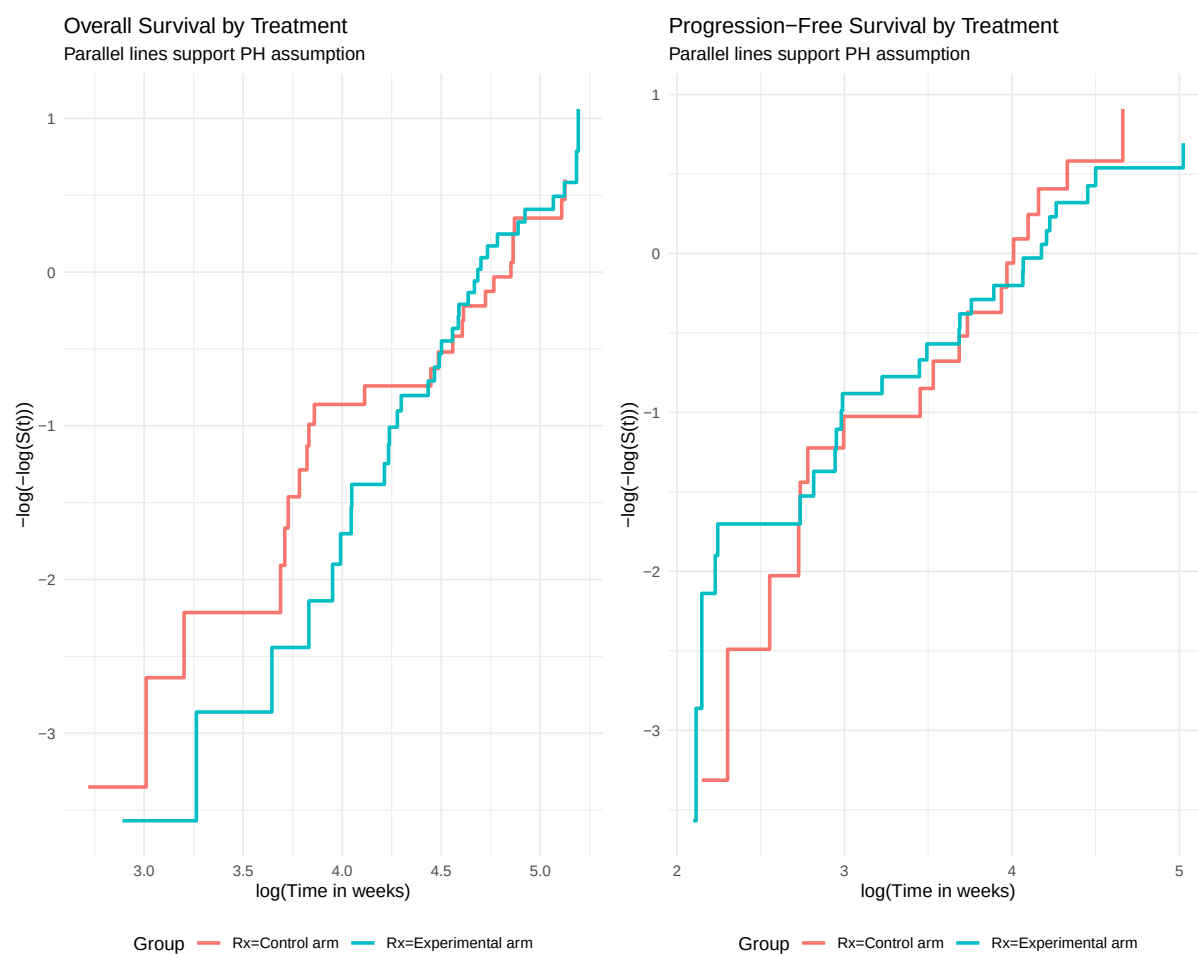
The proportional hazards assumption was tested using scaled Schoenfeld residuals. A significant p-value indicates potential violation of the PH assumption for that covariate.

Table 3: Schoenfeld Residual Test for Proportional Hazards Assumption

| Covariate       | <U+03C7><U+00B2> (OS) | p-value      | <U+03C7><U+00B2> (PFS) | p-value      |
|-----------------|-----------------------|--------------|------------------------|--------------|
| Age             | 1.85                  | 0.174        | 0.50                   | 0.480        |
| sex             | 0.10                  | 0.751        | 0.04                   | 0.847        |
| Rx              | 3.15                  | 0.076        | 0.04                   | 0.841        |
| crp_num         | 0.89                  | 0.345        | 3.22                   | 0.073        |
| tmb_braf_num    | 1.01                  | 0.315        | 5.33                   | 0.021        |
| Rx:crp_num      | 2.26                  | 0.133        | 1.21                   | 0.272        |
| Rx:tmb_braf_num | 0.57                  | 0.450        | 4.01                   | 0.045        |
| <b>GLOBAL</b>   | <b>7.32</b>           | <b>0.396</b> | <b>19.60</b>           | <b>0.007</b> |

**Note:** GLOBAL is the omnibus test for all covariates. Individual covariate tests help identify specific PH violations.

## 4.2 Log-Log Survival Plots



**Interpretation:** Approximately parallel lines on the log-log plot support the proportional hazards assumption. Crossing or converging lines suggest time-varying effects.

## 5 Primary Analysis: Firth-Corrected Cox Models

### 5.1 Standard Cox vs Firth

The standard Cox model is the unpenalized maximum-likelihood reference. Divergence between standard Cox and Firth estimates flags small-sample or near-separation instability that Firth corrects.

Table 4: Unified Model - Overall Survival: Standard Cox vs Firth

| Term                                      | Standard Cox HR (95% CI) | Firth HR (95% CI) |
|-------------------------------------------|--------------------------|-------------------|
| <b>Treatment x Biomarker Interactions</b> |                          |                   |
| CRP x Rx                                  | 0.88 (0.20-3.85)         | 0.78 (0.21-3.61)  |
| TMB/BRAF x Rx                             | 0.95 (0.30-2.97)         | 0.93 (0.30-2.85)  |
| <b>Main Effects and Covariates</b>        |                          |                   |
| Rx (Experimental)                         | 1.69 (0.79-3.63)         | 1.70 (0.81-3.67)  |
| CRP                                       | 0.45 (0.13-1.55)         | 0.52 (0.13-1.47)  |
| TMB/BRAF                                  | 0.91 (0.40-2.10)         | 0.94 (0.41-2.12)  |
| Age                                       | 1.00 (0.97-1.03)         | 1.00 (0.97-1.03)  |
| Sex                                       | 1.59 (0.90-2.81)         | 1.58 (0.90-2.79)  |

Table 5: Unified Model - Progression-Free Survival: Standard Cox vs Firth

| Term                                      | Standard Cox HR (95% CI) | Firth HR (95% CI) |
|-------------------------------------------|--------------------------|-------------------|
| <b>Treatment x Biomarker Interactions</b> |                          |                   |
| CRP x Rx                                  | 0.38 (0.06-2.34)         | 0.33 (0.07-2.15)  |
| TMB/BRAF x Rx                             | 0.61 (0.16-2.38)         | 0.60 (0.16-2.28)  |
| <b>Main Effects and Covariates</b>        |                          |                   |
| Rx (Experimental)                         | 2.04 (0.86-4.81)         | 2.01 (0.89-4.84)  |
| CRP                                       | 0.84 (0.18-3.95)         | 1.00 (0.19-3.60)  |
| TMB/BRAF                                  | 0.74 (0.27-1.99)         | 0.77 (0.28-2.01)  |
| Age                                       | 0.99 (0.95-1.02)         | 0.99 (0.96-1.02)  |
| Sex                                       | 1.21 (0.65-2.26)         | 1.20 (0.65-2.26)  |

### 5.2 Unified Model Results

The unified DAG-informed model includes CRP and TMB/BRAF as pre-treatment biomarkers with their treatment interactions, adjusted for Age and Sex.

**Formula:**  $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP:Rx} + \text{TMB/BRAF:Rx}$

#### 5.2.1 Overall Survival

Table 6: Unified Model: Overall Survival - Full Coefficient Table

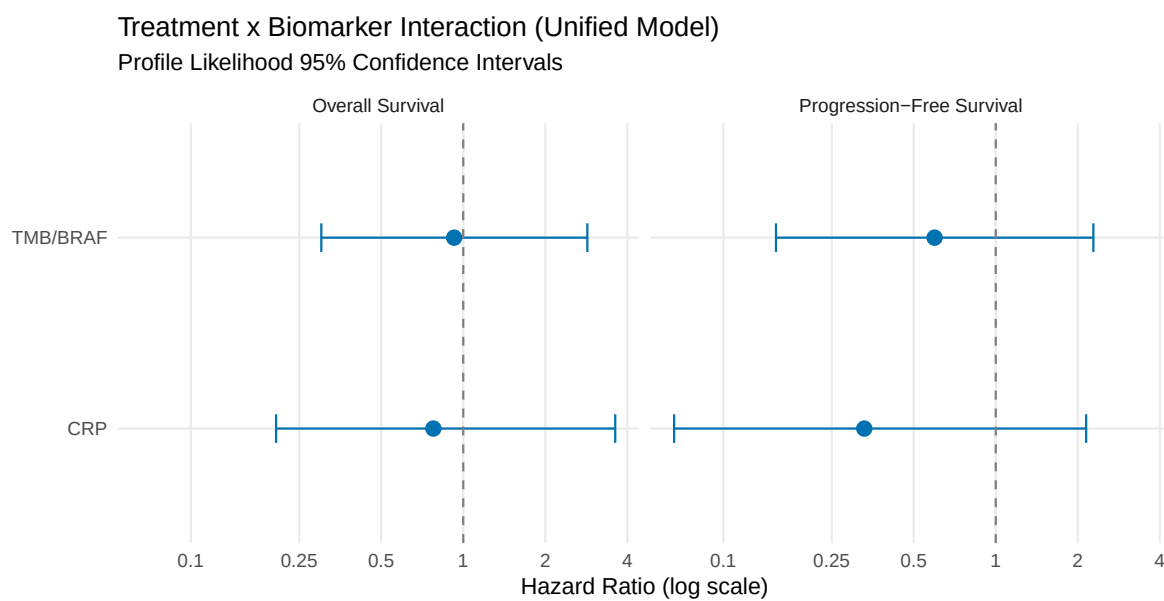
| Term                                      | HR (95% CI)      | PLRT p-value |
|-------------------------------------------|------------------|--------------|
| <b>Treatment x Biomarker Interactions</b> |                  |              |
| RxExperimental arm:crp_num                | 0.78 (0.21-3.61) | 0.728        |
| RxExperimental arm:tmb_braf_num           | 0.93 (0.30-2.85) | 0.893        |
| <b>Main Effects and Covariates</b>        |                  |              |
| Age                                       | 1.00 (0.97-1.03) | 0.987        |
| sex1                                      | 1.58 (0.90-2.79) | 0.110        |
| RxExperimental arm                        | 1.70 (0.81-3.67) | 0.164        |
| crp_num                                   | 0.52 (0.13-1.47) | 0.233        |
| tmb_braf_num                              | 0.94 (0.41-2.12) | 0.891        |

### 5.2.2 Progression-Free Survival

Table 7: Unified Model: Progression-Free Survival - Full Coefficient Table

| Term                                      | HR (95% CI)      | PLRT p-value |
|-------------------------------------------|------------------|--------------|
| <b>Treatment x Biomarker Interactions</b> |                  |              |
| RxExperimental arm:crp_num                | 0.33 (0.07-2.15) | 0.224        |
| RxExperimental arm:tmb_braf_num           | 0.60 (0.16-2.28) | 0.447        |
| <b>Main Effects and Covariates</b>        |                  |              |
| Age                                       | 0.99 (0.96-1.02) | 0.386        |
| sex1                                      | 1.20 (0.65-2.26) | 0.553        |
| RxExperimental arm                        | 2.01 (0.89-4.84) | 0.096        |
| crp_num                                   | 1.00 (0.19-3.60) | 0.998        |
| tmb_braf_num                              | 0.77 (0.28-2.01) | 0.594        |

### 5.3 Forest Plot



**Interpretation:**  $HR < 1$  indicates that biomarker-positive patients derive greater benefit from experimental treatment (reduced hazard of death/progression). The dashed line at  $HR = 1$  represents no differential treatment effect.

## 6 Sensitivity Analysis: Ridge Regression

Ridge regression (L2-penalized Cox,  $\alpha = 0$ ) shrinks all coefficients toward zero but does not eliminate any, providing a sensitivity check for the stability of the interaction estimates. The same unified model formula is used; only the main effects (CRP and TMB/BRAF) are penalized.

Table 8: Overall Survival: Firth vs Ridge Regression

| Interaction   | Firth (95% CI)   | Ridge (point est.) |
|---------------|------------------|--------------------|
| CRP x Rx      | 0.78 (0.21-3.61) | 0.39 (SE: 2.14)    |
| TMB/BRAF x Rx | 0.93 (0.30-2.85) | 0.87 (SE: 2.04)    |

*Note:* Ridge: L2 shrinkage on main effects. Substantial shrinkage vs Firth = estimate sensitive to regularization.

Table 9: Progression-Free Survival: Firth vs Ridge Regression

| Interaction   | Firth (95% CI)   | Ridge (point est.) |
|---------------|------------------|--------------------|
| CRP x Rx      | 0.33 (0.07-2.15) | 0.32 (SE: 2.15)    |
| TMB/BRAF x Rx | 0.60 (0.16-2.28) | 0.45 (SE: 2.05)    |

*Note:* Ridge applies L2 shrinkage to main effects only.

**Interpretation:** Firth provides the best unbiased point estimate given the sample size and data structure. Ridge applies L2 shrinkage toward zero (HR toward 1.0). Comparing the two methods reveals estimate stability: if Firth and Ridge agree closely, the estimate is robust; if Ridge substantially shrinks the interaction toward null, the effect is sensitive to regularization in this sample.

## 7 Exploratory: TLR as a Predictive Biomarker (Responder Analysis)

This section presents an **exploratory responder analysis** that includes TLR (Tumor Lesion Reduction) as an effect modifier in a Cox model with the same structural form as the primary analysis. Because TLR is measured post-randomization (at the first on-treatment CT scan), this is not a pre-treatment patient-selection analysis: TLR status is itself influenced by treatment, and the resulting TLR x Rx interaction is informative only as a hypothesis-generating, responder-stratified contrast. It cannot be used to guide treatment decisions at baseline. The primary causal-DAG-informed analysis remains the inferential anchor for predictive biomarker claims.

The model adjusts for the pre-treatment biomarkers **CRP** and **TMB/BRAF** as prognostic main effects (their treatment interactions belong to the primary analysis and are not re-estimated here). In the ridge sensitivity model, CRP and TMB/BRAF remain **unpenalized** (well-established prognostic adjustment covariates), while the **TLR main effect and TLR x Rx interaction are penalized**. These are the post-randomization terms whose stability we want to stress-test against the small effective sample size and the partially tautological TLR-PFS association.

**Formula:**  $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR:Rx}$

### 7.1 Descriptive Context

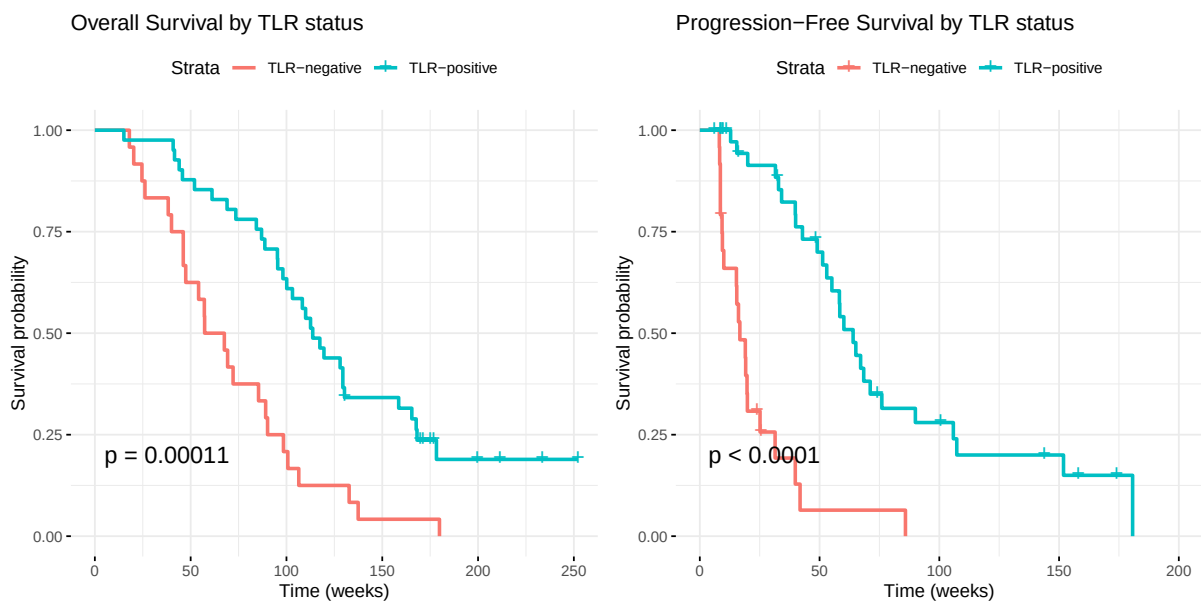
#### 7.1.1 TLR Prevalence by Treatment Arm

Table 10: TLR Prevalence by Treatment Arm (Descriptive)

| Treatment Arm            | N  | TLR-positive | %    |
|--------------------------|----|--------------|------|
| Control (FLOX)           | 29 | 22           | 75.9 |
| Experimental (FLOX/nivo) | 36 | 19           | 52.8 |

*Note:* TLR = Tumour Lesion Reduction  $\geq 10\%$  at first CT scan (post-treatment).

#### 7.1.2 Kaplan-Meier Curves Stratified by TLR



**Note:** These curves are purely descriptive and stratify by a post-treatment response variable.

The observed difference in survival by TLR status reflects treatment response, not a pre-treatment patient characteristic.

## 7.2 Standard Cox vs Firth

The standard Cox model is the unpenalized maximum-likelihood reference. Divergence between standard Cox and Firth estimates flags small-sample or near-separation instability that Firth corrects.

Table 11: TLR Responder Model - Overall Survival: Standard Cox vs Firth

| Term                         | Standard Cox HR (95% CI) | Firth HR (95% CI) |
|------------------------------|--------------------------|-------------------|
| <b>TLR Terms</b>             |                          |                   |
| TLR x Rx                     | 2.43 (0.74-7.95)         | 2.47 (0.75-7.75)  |
| TLR (main effect)            | 0.21 (0.08-0.54)         | 0.21 (0.08-0.55)  |
| <b>Adjustment Covariates</b> |                          |                   |
| Rx (Experimental)            | 0.60 (0.23-1.55)         | 0.59 (0.24-1.55)  |
| CRP                          | 0.52 (0.25-1.06)         | 0.53 (0.25-1.05)  |
| TMB/BRAF                     | 0.86 (0.48-1.52)         | 0.86 (0.48-1.52)  |
| Age                          | 1.01 (0.98-1.04)         | 1.01 (0.98-1.04)  |
| Sex                          | 1.48 (0.81-2.68)         | 1.47 (0.82-2.67)  |

Table 12: TLR Responder Model - Progression-Free Survival: Standard Cox vs Firth

| Term                         | Standard Cox HR (95% CI) | Firth HR (95% CI) |
|------------------------------|--------------------------|-------------------|
| <b>TLR Terms</b>             |                          |                   |
| TLR x Rx                     | 0.73 (0.19-2.76)         | 0.75 (0.20-2.66)  |
| TLR (main effect)            | 0.17 (0.06-0.50)         | 0.17 (0.06-0.52)  |
| <b>Adjustment Covariates</b> |                          |                   |
| Rx (Experimental)            | 1.41 (0.46-4.27)         | 1.35 (0.48-4.22)  |
| CRP                          | 0.41 (0.18-0.96)         | 0.43 (0.18-0.96)  |
| TMB/BRAF                     | 0.48 (0.24-0.99)         | 0.50 (0.24-0.99)  |
| Age                          | 1.00 (0.96-1.03)         | 1.00 (0.96-1.03)  |
| Sex                          | 1.23 (0.61-2.47)         | 1.22 (0.62-2.45)  |

## 7.3 Firth-Corrected Cox Model

### 7.3.1 Overall Survival

Table 13: TLR Responder Model: Overall Survival - Full Coefficient Table

| Term                                     | HR (95% CI)      | PLRT p-value |
|------------------------------------------|------------------|--------------|
| <b>Treatment x Biomarker Interaction</b> |                  |              |
| RxExperimental arm:tlr_num               | 2.47 (0.75-7.75) | 0.135        |
| <b>Main Effects and Covariates</b>       |                  |              |
| Age                                      | 1.01 (0.98-1.04) | 0.626        |
| sex1                                     | 1.47 (0.82-2.67) | 0.201        |
| RxExperimental arm                       | 0.59 (0.24-1.55) | 0.270        |
| crp_num                                  | 0.53 (0.25-1.05) | 0.071        |
| tmb_braf_num                             | 0.86 (0.48-1.52) | 0.609        |
| tlr_num                                  | 0.21 (0.08-0.55) | 0.002        |

### 7.3.2 Progression-Free Survival

Table 14: TLR Responder Model: Progression-Free Survival - Full Coefficient Table

| Term                                     | HR (95% CI)      | PLRT p-value |
|------------------------------------------|------------------|--------------|
| <b>Treatment x Biomarker Interaction</b> |                  |              |
| RxExperimental arm:tlr_num               | 0.75 (0.20-2.66) | 0.660        |
| <b>Main Effects and Covariates</b>       |                  |              |
| Age                                      | 1.00 (0.96-1.03) | 0.787        |
| sex1                                     | 1.22 (0.62-2.45) | 0.570        |
| RxExperimental arm                       | 1.35 (0.48-4.22) | 0.576        |
| crp_num                                  | 0.43 (0.18-0.96) | 0.039        |
| tmb_braf_num                             | 0.50 (0.24-0.99) | 0.046        |
| tlr_num                                  | 0.17 (0.06-0.52) | 0.003        |

#### 7.4 Ridge Sensitivity Analysis (TLR Terms Penalized)

Ridge regression is applied here with a targeted penalty: only the TLR main effect and the TLR x Rx interaction are L2-shrunk. CRP, TMB/BRAF, Age, sex, and Rx are estimated without penalty, since they enter the model as adjustment covariates whose effects are not the inferential target.

Table 15: TLR Responder Model - Overall Survival: Firth vs Ridge, all coefficients

| Term                                             | Firth (95% CI)   | Ridge HR (SE)   |
|--------------------------------------------------|------------------|-----------------|
| <b>TLR Terms (Ridge-penalized)</b>               |                  |                 |
| TLR x Rx                                         | 2.47 (0.75-7.75) | 2.12 (SE: 0.58) |
| TLR (main effect)                                | 0.21 (0.08-0.55) | 0.24 (SE: 0.47) |
| <b>Adjustment Covariates (Ridge-unpenalized)</b> |                  |                 |
| Age                                              | 1.01 (0.98-1.04) | 1.01 (SE: 0.02) |
| Sex                                              | 1.47 (0.82-2.67) | 1.49 (SE: 0.30) |
| Rx (Experimental)                                | 2.47 (0.75-7.75) | 0.66 (SE: 0.48) |
| CRP                                              | 0.53 (0.25-1.05) | 0.52 (SE: 0.36) |
| TMB/BRAF                                         | 0.86 (0.48-1.52) | 0.86 (SE: 0.29) |

*Note:* Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/B

Table 16: TLR Responder Model - Progression-Free Survival: Firth vs Ridge, all coefficients

| Term                                             | Firth (95% CI)   | Ridge HR (SE)   |
|--------------------------------------------------|------------------|-----------------|
| <b>TLR Terms (Ridge-penalized)</b>               |                  |                 |
| TLR x Rx                                         | 0.75 (0.20-2.66) | 0.68 (SE: 0.64) |
| TLR (main effect)                                | 0.17 (0.06-0.52) | 0.19 (SE: 0.53) |
| <b>Adjustment Covariates (Ridge-unpenalized)</b> |                  |                 |
| Age                                              | 1.00 (0.96-1.03) | 0.99 (SE: 0.02) |
| Sex                                              | 1.22 (0.62-2.45) | 1.24 (SE: 0.35) |
| Rx (Experimental)                                | 0.75 (0.20-2.66) | 1.49 (SE: 0.55) |
| CRP                                              | 0.43 (0.18-0.96) | 0.41 (SE: 0.43) |
| TMB/BRAF                                         | 0.50 (0.24-0.99) | 0.49 (SE: 0.36) |

*Note:* Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/B

**Interpretation:** The Firth HR is the best unbiased point estimate of the TLR x Rx interaction under penalized maximum likelihood. The ridge HR applies L2 shrinkage targeted at the TLR terms; if the ridge estimate is substantially closer to HR = 1.0 than the Firth estimate, the responder-stratified interaction is sensitive to regularization, consistent with a small-sample, partly tautological signal. Findings here are exploratory and should not be used for pre-treatment patient selection, since TLR is not measurable at baseline. They may motivate landmark or formal causal-mediation analyses in future work.

## 8 Discussion

### 8.1 Primary interaction results

Neither biomarker-treatment interaction reached conventional significance in the unified model. For **overall survival**,  $\text{CRP} \times \text{Rx}$  yielded HR 0.78 (95% CI 0.21–3.61, PLRT  $p = 0.728$ ) and  $\text{TMB/BRAF} \times \text{Rx}$  HR 0.93 (0.30–2.85,  $p = 0.893$ ) — point estimates close to null with very wide confidence intervals. For **progression-free survival**, the  $\text{CRP} \times \text{Rx}$  interaction showed a more directionally suggestive effect (HR 0.33, 0.07–2.15,  $p = 0.224$ ): CRP-positive patients in the experimental arm had roughly one-third the hazard of progression relative to CRP-positive controls, though the CI spans more than an order of magnitude. These results are consistent with the trial being underpowered to detect treatment-effect heterogeneity ( $n = 65$ , 48 progressions, 56 deaths): even a moderate subgroup effect (HR  $\sim 0.5$ ) would require far larger samples for reliable estimation.

The PFS pattern is worth noting as a hypothesis-generating finding. The CRP direction (HR 0.33) is consistent with the DAG association test showing  $\text{CRP} \rightarrow \text{PFS}$  (HR 0.40,  $p = 0.004$ ) and the biological hypothesis that low CRP (reflecting lower systemic inflammation) identifies patients more likely to respond to immunotherapy. However, the OS CRP interaction estimate (HR 0.78) is considerably closer to null, suggesting that any PFS benefit does not clearly translate to an OS benefit in this dataset.

### 8.2 Proportional hazards and TMB/BRAF

The PH assumption was well supported in overall survival (Schoenfeld global  $p = 0.40$ ). In progression-free survival, however, the global test was significant ( $p = 0.007$ ), driven mainly by TMB/BRAF — both the main effect ( $p = 0.021$ ) and the  $\text{TMB/BRAF} \times \text{Rx}$  interaction ( $p = 0.045$ ) violated the PH assumption. This indicates that the hazard ratio for TMB/BRAF (and its modification by treatment) is not constant over time in PFS, which may reflect different kinetics of early vs late events in TMB/BRAF-positive tumours. The PFS  $\text{TMB/BRAF} \times \text{Rx}$  interaction estimate (HR 0.60) should therefore be interpreted with caution; a time-varying Cox model or landmark analysis would be more appropriate for that specific question.

### 8.3 Ridge regression sensitivity

For OS, Ridge shrinks the  $\text{CRP} \times \text{Rx}$  estimate substantially toward null (Firth HR 0.78  $\rightarrow$  Ridge HR 0.39), suggesting the OS estimate is sensitive to regularization and should be treated as imprecise. The  $\text{TMB/BRAF} \times \text{Rx}$  OS estimate shows minimal shrinkage (0.93  $\rightarrow$  0.87), consistent with a near-null interaction. For PFS, the  $\text{CRP} \times \text{Rx}$  estimate is essentially unchanged by Ridge (Firth 0.33  $\rightarrow$  Ridge 0.32), indicating that the directional PFS finding is robust to regularization — the data support it even when the model is penalized for large coefficients.  $\text{TMB/BRAF} \times \text{Rx}$  PFS shows moderate shrinkage (0.60  $\rightarrow$  0.45).

### 8.4 TLR responder analysis (exploratory)

The exploratory responder model —  $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR}:\text{Rx}$  — estimates the differential treatment effect between TLR-positive and TLR-negative subgroups while adjusting for the pre-treatment biomarkers as prognostic main effects. The Firth-estimated  $\text{TLR} \times \text{Rx}$  HR was 2.47 (0.75–7.75) for overall survival and 0.75 (0.20–2.66) for progression-free survival. The OS estimate sits **above 1.0**, which directionally implies the (TLR-positive vs TLR-negative) hazard contrast is *worse* on the experimental arm than on the control arm — the opposite of a “TLR-positive predicts immunotherapy benefit” pattern. PFS is closer to null. Ridge shrinkage of the TLR terms gives 2.12 (SE: 0.58) (OS) and 0.68 (SE: 0.64) (PFS): ridge moves the OS estimate modestly toward null but preserves the directional pattern,

and barely changes the PFS estimate. PLRT p-values (0.135 OS, 0.660 PFS) do not reach conventional significance, consistent with the wide profile-likelihood CIs in this small sample.

The directional pattern is consistent with the descriptive imbalance: TLR-positive prevalence is 75.9% in the control arm and 52.8% in the experimental arm. If TLR cleanly captured immunotherapy response one would expect higher TLR-positivity in the experimental arm, not lower. The observed prevalence pattern, together with the OS direction ( $HR > 1$ ) of the  $TLR \times Rx$  interaction, is more consistent with TLR acting as a **prognostic** marker (identifying chemotherapy-responsive disease, which is more prevalent in the control arm by chance or by timing) than as a predictive marker for immunotherapy benefit. Possible explanations for the imbalance include small-sample randomization imbalance, a timing artefact (TLR measured at a fixed CT that may occur during different treatment phases across arms), or genuine heterogeneity in the patient mix. Because TLR is itself influenced by treatment, the  $TLR \times Rx$  interaction is a responder-stratified contrast — useful for hypothesis generation, but not a causal predictive-biomarker estimate. Clinical decision support would require a formal causal mediation or landmark analysis.

## 9 Summary

**Study:** METIMMOX-1 biomarker subgroup analysis. N = 65 complete cases, 56 OS events, 48 PFS events.

**Methods:** Two Firth-corrected Cox analyses are presented. **Primary (DAG-informed):**  $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP}:\text{Rx} + \text{TMB/BRAF}:\text{Rx}$  — pre-treatment biomarkers as effect modifiers; TLR omitted because it is post-randomization. **Exploratory responder analysis:**  $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR}:\text{Rx}$  — adds TLR and its treatment interaction, with CRP and TMB/BRAF retained as prognostic main-effect adjustments. Profile likelihood CIs, PLRT for interaction testing. Ridge sensitivity analysis penalizes the CRP/TMB main effects in the primary model and the TLR terms (main effect + TLR:Rx) in the exploratory model.

**Proportional hazards:** No violations in OS (global  $p = 0.40$ ). In PFS, the TMB/BRAF main effect ( $p = 0.021$ ) and TMB/BRAF  $\times$  Rx interaction ( $p = 0.045$ ) violate the PH assumption; PFS TMB/BRAF estimates should be interpreted with caution.

### Interaction estimates:

| Interaction          | OS HR (95% CI)   | PLRT p | PFS HR (95% CI)  | PLRT p |
|----------------------|------------------|--------|------------------|--------|
| CRP $\times$ Rx      | 0.78 (0.21–3.61) | 0.728  | 0.33 (0.07–2.15) | 0.224  |
| TMB/BRAF $\times$ Rx | 0.93 (0.30–2.85) | 0.893  | 0.60 (0.16–2.28) | 0.447  |

No interaction is statistically significant. CRP  $\times$  Rx PFS (HR 0.33) is the most directionally consistent finding and is robust to Ridge regularization (Ridge HR 0.32). TMB/BRAF OS shows near-null interaction (Ridge HR 0.87 Firth); TMB/BRAF PFS violated PH.

**TLR responder analysis (exploratory):** Firth TLR  $\times$  Rx HR 2.47 (0.75-7.75) for OS and 0.75 (0.20-2.66) for PFS. Ridge-penalized TLR terms: 2.12 (SE: 0.58) (OS), 0.68 (SE: 0.64) (PFS). PLRT p-values (0.135 OS, 0.660 PFS) do not reach conventional significance in this small sample. TLR-positive prevalence: control 75.9%, experimental 52.8% — markedly higher TLR-positivity in the control arm. The OS direction (HR  $> 1$ ) of TLR  $\times$  Rx, combined with the prevalence pattern, is more consistent with TLR acting as a prognostic marker for chemo-responsive disease than as a predictive marker for immunotherapy benefit. Because TLR is post-randomization, this is a responder-stratified, hypothesis-generating contrast rather than a causal predictive-biomarker estimate.

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